The sharp norm asymptotic of the orthogonal cycle projection

Candidate full answer to the orthogonal branch of Question 32 in arXiv:2305.12582

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Status. Candidate full solution of the P_{orth} branch, subject to human verification. If P_{orth} is the orthogonal projection from the edge space of the discrete two-dimensional torus onto its cycle space, then

$$\|P_{\mathrm{orth}}\|_{\ell_1(E_n)\to\ell_1(E_n)} = \frac{4}{\pi}\log n + O(1).$$

The leading constant is sharp. The separate $P_{\min}$ branch remains open.

1 Source and exact question

The source is S. J. Dilworth, D. Kutzarova, and M. I. Ostrovskii, *Cycle spaces: invariant projections and applications to transportation cost* [1], arXiv:2305.12582. For the graph $\mathbb{T}_n = \mathbb{Z}_n \times \mathbb{Z}_n$, let E_n consist of the positively oriented horizontal and vertical edges, let Z_n be its cycle space, and let $P_{\mathrm{orth}} : \ell_2(E_n) \to Z_n$ be the orthogonal projection, regarded also as an operator on $\ell_1(E_n)$.

After proving only

$$a_1\sqrt{\log n} \le \|P_{\min}\|_1 \le a_2 \log n,$$

the authors state Question 32 on source PDF page 18: "Find good asymptotic estimates (improving on Proposition 31) for $\|P_{\min}\|_1$ and/or $\|P_{\mathrm{orth}}\|_1$." Figure 1 shows the statement and its immediate context.

The theorem below gives a sharp answer to the orthogonal-projection branch. It does not identify $P_{\min}$, which the source shows already differs from P_{orth} when $n = 6$.

2 Idea of the proof

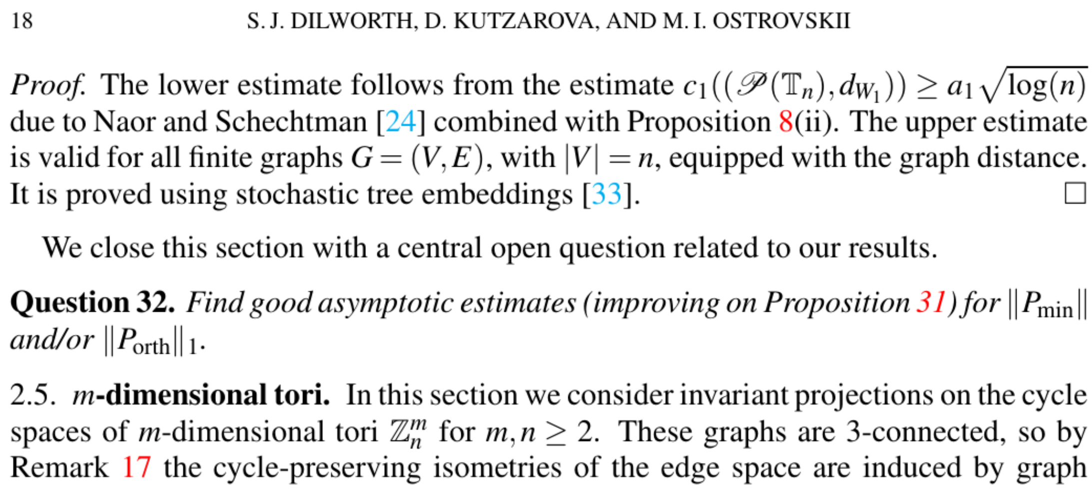
The orthogonal complement of the cycle space is the cut, or gradient, space. Thus the complementary projection is the discrete Helmholtz projection

$$Q_n = I - P_{\mathrm{orth}} = d\Delta^\dagger d^*,$$

where d is the forward difference, $\Delta = d^*d$, and the dagger denotes the inverse on mean-zero vertex functions. Translation and coordinate symmetry reduce its ℓ_1 operator norm to the absolute sum of one column. That column is a discrete dipole field: the two relevant kernels are second differences of the Green function.

At distance r from the pole, the Fourier symbol has the same singular part as $\xi_j\xi_1/|\xi|^2$. Its inverse Fourier transform is

$$H_{11}(x) = -\frac{\cos(2\theta_x)}{2\pi r^2}, \qquad H_{21}(x) = -\frac{\sin(2\theta_x)}{2\pi r^2}.$$



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Proof. The lower estimate follows from the estimate $c_1((\mathscr{P}(\mathbb{T}_n), d_{W_1})) \geq a_1\sqrt{\log(n)}$ due to Naor and Schechtman [24] combined with Proposition 8(ii). The upper estimate is valid for all finite graphs $G = (V,E)$, with $|V| = n$, equipped with the graph distance. It is proved using stochastic tree embeddings [33]. □

We close this section with a central open question related to our results.

Question 32. *Find good asymptotic estimates (improving on Proposition 31) for* $\|P_{\min}\|_1$ *and/or* $\|P_{\mathrm{orth}}\|_1$.

2.5. ***m*-dimensional tori.** In this section we consider invariant projections on the cycle spaces of m-dimensional tori $\mathbb{Z}_n^m$ for $m, n \geq 2$. These graphs are 3-connected, so by Remark 17 the cycle-preserving isometries of the edge space are induced by graph

Figure 1: Question 32 on page 18 of the arXiv source [1].

A singular-symbol estimate makes the error $O(r^{-3})$. Passing from the infinite lattice to the finite torus periodizes H; square symmetry cancels the degree -2 tail, leaving a bounded scaled periodic kernel. Hence all errors have bounded total ℓ_1 mass.

The main term contributes

$$\frac{1}{2\pi}\int_1^n \frac{dr}{r}\int_0^{2\pi}(|\cos 2\theta| + |\sin 2\theta|)\,d\theta = \frac{4}{\pi}\log n + O(1).$$

Finally, a trace computation shows that replacing Q_n by P_{orth} adds exactly n^{-2} to the norm.

3 Fourier kernel estimates

Write $e_1 = (1,0)$, $e_2 = (0,1)$ and

$$(df)_j(x) = f(x+e_j) - f(x), \qquad \Delta = d^*d.$$

For $\vartheta \in (-\pi,\pi]^2$, put

$$b_j(\vartheta) = e^{i\vartheta_j} - 1, \qquad \lambda(\vartheta) = |b_1(\vartheta)|^2 + |b_2(\vartheta)|^2,$$

and, away from the origin,

$$m_j(\vartheta) = \frac{b_j(\vartheta)\overline{b_1(\vartheta)}}{\lambda(\vartheta)} \qquad (j = 1,2).$$

The value assigned at 0 will never matter beyond an $O(n^{-2})$ term.

For $x \neq 0$, define the homogeneous kernels

$$H_j(x) = \frac{1}{2\pi}\left(\frac{\delta_{j1}}{|x|^2} - \frac{2x_jx_1}{|x|^4}\right). \tag{1}$$

Lemma 1 (Infinite-lattice singular symbol)**.** *Let*

$$\widehat{m}_j(a) = \frac{1}{(2\pi)^2}\int_{(-\pi,\pi]^2} e^{ia\cdot\vartheta} m_j(\vartheta)\,d\vartheta, \qquad a \in \mathbb{Z}^2.$$

Then, uniformly for $a \neq 0$,

$$\widehat{m}_j(a) = H_j(a) + O((1+|a|)^{-3}).$$

Proof. Taylor expansion at the unique singular point gives

$$m_j(\vartheta) = \frac{\vartheta_j \vartheta_1}{|\vartheta|^2} + r_j(\vartheta), \qquad |\partial^\alpha r_j(\vartheta)| \le C_\alpha |\vartheta|^{1-|\alpha|}. \tag{2}$$

Choose a smooth cutoff χ supported near zero and equal to one in a smaller neighborhood. The part of m_j away from zero is smooth on the Fourier torus, so repeated integration by parts makes its coefficients $O_N(|a|^{-N})$.

The inverse Fourier transform on $\mathbb{R}^2$ of $\vartheta_j \vartheta_1/|\vartheta|^2$ is precisely H_j: this follows by differentiating the fundamental solution $-(2\pi)^{-1} \log|x|$ of the positive Laplacian. Multiplication by χ convolves H_j with a Schwartz function of integral one. Since $|\nabla H_j(x)| \le C|x|^{-3}$, splitting that convolution into $|y| \le |a|/2$ and its Schwartz tail shows that the result is $H_j(a) + O(|a|^{-3})$.

It remains to estimate χr_j. A smooth dyadic decomposition into frequency annuli of radius ρ turns (2) into the standard bound

$$|\mathcal{F}^{-1}(\chi r_j)_\rho(a)| \le C_N \rho^3 (1 + \rho|a|)^{-N}.$$

Summing separately over $\rho \le |a|^{-1}$ and $\rho > |a|^{-1}$ gives $O(|a|^{-3})$. Combining the three pieces proves the assertion. □

For $x \in \mathbb{T}_n$, choose its representative in $[-n/2, n/2)^2$ and write $\rho_n(x) = |x|$.

Lemma 2 (Torus dipole kernel)**.** *Define*

$$K_{j,n}(x) = \frac{1}{n^2} \sum_{k \in \mathbb{Z}_n^2 \setminus \{0\}} e^{2\pi i k \cdot x/n} m_j(2\pi k/n). \tag{3}$$

There is an absolute constant C such that

$$K_{j,n}(x) = H_j(x) + O(\rho_n(x)^{-3} + n^{-2}) \qquad (1 \le \rho_n(x) \le n/8), \tag{4}$$

$$|K_{j,n}(x)| \le C n^{-2} \qquad (\rho_n(x) > n/8). \tag{5}$$

Proof. We include the periodization step because it is the point at which a non-summable absolute $|a|^{-2}$ estimate would lose the theorem. Fourier aliasing, first for Abel-regularized Fourier series and then by letting the regularization parameter tend to zero, gives

$$K_{j,n}(x) = \sum_{\ell \in \mathbb{Z}^2}^{\text{sym}} \widehat{m}_j(x + n\ell) + O(n^{-2}). \tag{6}$$

The last term accounts for the arbitrary value of the multiplier at the single grid point 0 and for the bounded change from Abel to centered-square summation of the homogeneous degree -2 term. Lemma 1 makes the periodization of its remainder absolutely convergent.

For the main term, define on $\mathbb{R}^2 \setminus \mathbb{Z}^2$

$$\mathcal{H}_j(z) = \lim_{N \to \infty} \sum_{\|\ell\|_\infty \le N} H_j(z + \ell).$$

This symmetric limit exists. Indeed, $H_1(\ell) = (\ell_2^2 - \ell_1^2)/(2\pi|\ell|^4)$, so its sum over every centered square of nonzero lattice points vanishes under coordinate interchange. Likewise $H_2(\ell) = -\ell_1\ell_2/(\pi|\ell|^4)$, whose sum vanishes under either coordinate reflection. Consequently,

$$\mathcal{H}_j(z) - H_j(z) = \sum_{\ell \neq 0} [H_j(z+\ell) - H_j(\ell)],$$

and this series converges locally uniformly because its tail is $O(|z||\ell|^{-3})$. Thus $\mathcal{H}_j(z) - H_j(z)$ is bounded for $|z| \le 1/8$, while $\mathcal{H}_j$ itself is bounded on the compact part of the unit torus at distance at least $1/8$ from the pole.

Homogeneity now gives

$$\sum_{\ell}^{\mathrm{sym}} H_j(x + n\ell) = n^{-2}\mathcal{H}_j(x/n).$$

For $|x| \le n/8$, this is $H_j(x) + O(n^{-2})$; away from that disk it is $O(n^{-2})$. The absolutely convergent error from Lemma 1 is $O(|x|^{-3} + n^{-3})$ in the first region and $O(n^{-3})$ in the second. Substitution in (6) proves (4)–(5). $\square$

4 Sharp norm asymptotic

Theorem 3. *For the two-dimensional discrete torus $\mathbb{T}_n = \mathbb{Z}_n^2$,*

$$\boxed{\|P_{\mathrm{orth}}\|_{\ell_1(E_n)\to\ell_1(E_n)} = \frac{4}{\pi}\log n + O(1).}$$

In particular, $\|P_{\mathrm{orth}}\|_1 = \Theta(\log n)$.

Proof. The cycle space is $\ker d^*$ and the cut space is $\operatorname{ran} d$. It follows from the orthogonal decomposition of the edge space that

$$Q_n := I - P_{\mathrm{orth}} = d\Delta^{\dagger}d^*.$$

Under the discrete Fourier transform, the 2×2 multiplier of Q_n is

$$\left(\frac{b_j(\vartheta)\overline{b_\ell(\vartheta)}}{\lambda(\vartheta)}\right)_{j,\ell=1}^{2}$$

at nonzero frequencies and zero at frequency zero. Hence, up to harmless translations caused by the edge orientation, the column of Q_n at the horizontal edge based at zero has components $K_{1,n}$ and $K_{2,n}$. All columns have the same absolute sum by translation and coordinate symmetry, so

$$\|Q_n\|_1 = \sum_{x\in\mathbb{T}_n} (|K_{1,n}(x)| + |K_{2,n}(x)|). \tag{7}$$

The contribution of $x = 0$ is bounded because the multiplier matrix is an orthogonal projection at every frequency. Lemma 2 shows that the region $\rho_n(x) > n/8$ also contributes $O(1)$, and in the remaining region the total difference between the summands in (7) and $|H_1(x)| + |H_2(x)|$ is bounded by

$$C \sum_{1\le|x|\le n/8} (|x|^{-3} + n^{-2}) = O(1).$$

Writing $x = r(\cos\theta, \sin\theta)$ in (1) gives

$$|H_1(x)| + |H_2(x)| = \frac{|\cos 2\theta| + |\sin 2\theta|}{2\pi r^2}.$$

The sum of this locally Lipschitz homogeneous function over lattice points differs from its integral over the corresponding union of unit squares by $O(1)$: the error on a square at radius r is $O(r^{-3})$, and those errors are summable over annuli. Therefore

$$\begin{aligned}\|Q_n\|_1 &= \frac{1}{2\pi}\int_1^{n/8}\frac{dr}{r}\int_0^{2\pi}(|\cos 2\theta| + |\sin 2\theta|)\,d\theta + O(1)\\ &= \frac{1}{2\pi}(4+4)\log n + O(1) = \frac{4}{\pi}\log n + O(1).\end{aligned}$$

It remains only to pass from Q_n to P_{orth}. Edge transitivity and $\operatorname{rank} Q_n = n^2 - 1$ imply that every diagonal entry of Q_n is

$$q_{ee} = \frac{\operatorname{trace} Q_n}{|E_n|} = \frac{n^2-1}{2n^2}.$$

The off-diagonal entries of $P_{\text{orth}} = I - Q_n$ are the negatives of those of Q_n, while the two diagonal entries in the selected columns are positive. Thus

$$\|P_{\text{orth}}\|_1 - \|Q_n\|_1 = (1 - q_{ee}) - q_{ee} = \frac{1}{n^2}.$$

The asserted formula follows. □

5 Verification, limitations, and novelty

1. **Exact finite Fourier check.** The script `code/verify_fft.py` evaluates the multiplier in (3). It reproduces the source values

$$\frac{19}{9},\quad \frac{41}{16},\quad \frac{69}{25},\quad \frac{3839}{1260}$$

for $n = 3, 4, 5, 6$. For $n = 64, 128, 256, 512$, the residuals $\|P_{\text{orth}}\|_1 - (4/\pi)\log n$ are approximately $0.71556, 0.71351, 0.71268, 0.71234$. This computation is not used as proof.

2. **Sensitive analytic point.** The leading lattice kernel is not absolutely summable in two dimensions. Lemma 2 therefore uses symmetric/Abel periodization, not a naive absolute periodization. The cancellation of $H_j(\ell)$ on centered squares and the absolutely convergent difference $H_j(z+\ell) - H_j(\ell)$ are the key checks.

3. **Scope.** The result completely determines the first-order asymptotic of P_{orth} and hence answers the "and/or P_{orth}" branch of Question 32. It does not determine $P_{\min}$ or its norm.

4. **Bounded novelty check.** The run registry, solution, attempt, and proof-gap indexes were searched for arXiv:2305.12582 and the core projection/torus terms. Web searches through 9 August 2026 used the exact title and question, "orthogonal cycle projection", "discrete torus Green function", and the proposed "$4/\pi \log n$" formula. No later answer was found. The published author offprint still contains the question as Question 32. This supports, but does not certify, novelty.

5. **Human-review recommendation.** Verify Lemma 2, especially the Abel-regularized aliasing identity and the uniform boundedness of $\mathcal{H}_j$ away from its pole. Then check the Fourier orientation only up to translation; the absolute column sum is invariant under those shifts.

References

[1] S. J. Dilworth, D. Kutzarova, and M. I. Ostrovskii, *Cycle spaces: invariant projections and applications to transportation cost*, arXiv:2305.12582 (2023), Question 32.